An Information-Theoretic Verification Proof of the Multinomial Kelly Rule

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Abstract

We revisit the classical single-event multinomial Kelly problem from an information-theoretic point of view. The core observation is that once the active set is identified, the optimal state-contingent wealth profile is the Arrow–Debreu likelihood ratio between the bettor's subjective law and a coarse-grained state-price law. This suggests a particularly short global proof: one postulates the corresponding wealth profile, then verifies its optimality for all feasible portfolios by the elementary inequality $\log x \leq x - 1$. The support is determined by conditional favorability relative to a reserve bucket, and after sorting by edge the optimal support is a prefix. We also record the complementary Kullback–Leibler refinement identity, which explains the same support rule as a one-step information gain. The theorem itself is classical; the contribution here is a tight, self-contained proof that foregrounds the information-theoretic and state-price structure.

1 Introduction

Kelly's original paper linked optimal capital growth to information rate and thereby gave logarithmic betting a distinctly information-theoretic interpretation [2]. In the single-event multinomial setting, the corresponding optimization problem is classical: one distributes wealth across mutually exclusive outcomes so as to maximize expected log terminal wealth. Explicit formulas and support-selection rules are known in the horse-race and pari-mutuel literature; see, for example, Rosner [3] and Smoczynski and Tomkins [4]. More recently, Bleuler, Lapidoth and Pfister [1] emphasized again that gambling problems often admit clean divergence-based proofs.

The purpose of this note is not to claim a new theorem. Rather, it isolates a particularly concise proof architecture for the fixed-odds multinomial Kelly rule. The proof has two ingredients.

First, on any prescribed proper support, the problem is naturally a state-price optimization over terminal wealth levels. A one-line Gibbs-inequality argument shows that the expected-log-optimal wealth profile is the Arrow–Debreu likelihood ratio between the bettor's subjective coarse-grained law and the corresponding coarse-grained implied-probability law. This yields the familiar closed-form Kelly fractions.

Second, this fixed-support calculation suggests the correct global candidate wealth profile. Once that candidate is written down, global optimality follows from a verification argument based on the elementary inequality $\log x \leq x - 1$. The resulting proof is short, conceptual, and entirely information-theoretic in spirit. The conditional-favorability threshold emerges as the exact penalty term for placing wealth on excluded outcomes. We also record an alternative Kullback–Leibler refinement identity, which explains the same support rule as a one-step information gain.

2 Model and notation

Consider a single event with mutually exclusive outcomes $i = 1, \ldots, n$. The bettor's probabilities are

$$p_i > 0, \qquad \sum_{i=1}^{n} p_i = 1,$$

and the gross decimal odds are $d_i > 0$. It is convenient to write

$$\hat{p}_i := \frac{1}{d_i}, \qquad s_i := \frac{p_i}{\hat{p}_i} = p_i d_i.$$

Thus $\hat{p}_i$ is the state price of one unit of wealth delivered in state i, while s_i is the usual edge ratio.

A portfolio is a vector $f = (f_1, \ldots, f_n)$ satisfying

$$f_i \geq 0, \qquad F := \sum_{i=1}^{n} f_i \leq 1.$$

If outcome i occurs, terminal wealth is

$$W_i(f) = 1 - F + d_i f_i.$$

The Kelly objective is

$$G(f) := \sum_{i=1}^{n} p_i \log W_i(f).$$

For a set $A \subseteq \{1, \ldots, n\}$, let

$$P_A := \sum_{i \in A} p_i, \qquad \hat{P}_A := \sum_{i \in A} \hat{p}_i, \qquad Q_A := 1 - P_A, \qquad \hat{Q}_A := 1 - \hat{P}_A.$$

When A is interpreted as the active set, Q_A and $\hat{Q}_A$ are the subjective and implied masses of the complementary reserve bucket.

Definition 2.1. For a proper set $A \subsetneq \{1, \ldots, n\}$ with $\hat{Q}_A > 0$, define its *reserve ratio*

$$\rho(A) := \frac{Q_A}{\hat{Q}_A}.$$

An outcome $k \notin A$ is *conditionally favorable relative to* A if

$$s_k > \rho(A).$$

Equivalently,

$$\frac{p_k}{Q_A} > \frac{\hat{p}_k}{\hat{Q}_A}.$$

The second form makes the meaning clear: after all inactive outcomes are collapsed into one reserve bucket, outcome k is favorable exactly when its conditional subjective probability exceeds its conditional implied probability.

Throughout, we sort the outcomes so that

$$s_1 \geq s_2 \geq \cdots \geq s_n.$$

Write $A_m := \{1, \ldots, m\}$ for the prefix of length m, with the convention $A_0 = \varnothing$.

3 A state-price lemma and the fixed-support problem

The following elementary optimization lemma is the engine behind the fixed-support calculation and the motivation for the global candidate.

Lemma 3.1 (Gibbs / state-price lemma)**.** *Let $\mathcal{X}$ be finite. Let $\alpha = (\alpha_x)_{x\in\mathcal{X}}$ be a probability vector and let $\beta = (\beta_x)_{x\in\mathcal{X}}$ be a strictly positive probability vector. Then*

$$\sup\Big\{\sum_{x\in\mathcal{X}} \alpha_x \log g_x : \ g_x > 0, \ \sum_{x\in\mathcal{X}} \beta_x g_x = 1\Big\} = \sum_{x\in\mathcal{X}} \alpha_x \log \frac{\alpha_x}{\beta_x} = D_{\mathrm{KL}}(\alpha\|\beta),$$

and the unique maximizer is

$$g_x^* = \frac{\alpha_x}{\beta_x} \qquad (x \in \mathcal{X}).$$

Proof. Write

$$\sum_x \alpha_x \log g_x = \sum_x \alpha_x \log\Big(\frac{\beta_x g_x}{\alpha_x}\Big) + \sum_x \alpha_x \log\Big(\frac{\alpha_x}{\beta_x}\Big).$$

Because log is concave and $\sum_x \alpha_x = 1$, Jensen's inequality gives

$$\sum_x \alpha_x \log\Big(\frac{\beta_x g_x}{\alpha_x}\Big) \le \log\Big(\sum_x \beta_x g_x\Big) = 0.$$

Hence

$$\sum_x \alpha_x \log g_x \le \sum_x \alpha_x \log \frac{\alpha_x}{\beta_x}.$$

Equality in Jensen holds if and only if $\beta_x g_x/\alpha_x$ is constant in x; the budget constraint forces that constant to be 1, so $g_x = \alpha_x/\beta_x$ for every x. □

Proposition 3.2 (Fixed-support formula)**.** *Let $A \subsetneq \{1,\dots,n\}$ be nonempty, and assume $\hat{Q}_A > 0$. Among portfolios satisfying*

$$f_i > 0 \ (i \in A), \qquad f_j = 0 \ (j \notin A),$$

the unique interior maximizer is

$$f_i^A = p_i - \hat{p}_i \frac{Q_A}{\hat{Q}_A} \qquad (i \in A), \tag{1}$$

with reserve fraction

$$r_A := 1 - \sum_{i=1}^n f_i^A = \frac{Q_A}{\hat{Q}_A}. \tag{2}$$

The associated terminal wealth levels are

$$W_i(f^A) = \frac{p_i}{\hat{p}_i} \qquad (i \in A), \qquad W_j(f^A) = \frac{Q_A}{\hat{Q}_A} \qquad (j \notin A),$$

and the optimized value is

$$V(A) := G(f^A) = \sum_{i\in A} p_i \log \frac{p_i}{\hat{p}_i} + Q_A \log \frac{Q_A}{\hat{Q}_A} = D_{\mathrm{KL}}(\alpha^A\|\beta^A), \tag{3}$$

where

$$\alpha^A = (p_i)_{i\in A} \cup \{Q_A\}, \qquad \beta^A = (\hat{p}_i)_{i\in A} \cup \{\hat{Q}_A\}.$$

Finally,

$$f_i^A > 0 \iff \frac{p_i}{\hat{p}_i} > \frac{Q_A}{\hat{Q}_A} \iff s_i > \rho(A). \tag{4}$$

Proof. Write $r := 1 - F$ and define terminal-wealth variables

$$g_i := r + d_i f_i \quad (i \in A), \qquad g_0 := r.$$

Then

$$G(f) = \sum_{i \in A} p_i \log g_i + Q_A \log g_0,$$

and the budget identity is

$$\sum_{i \in A} \hat{p}_i g_i + \hat{Q}_A g_0 = 1.$$

Apply Lemma 3.1 on the augmented state space $A \cup \{0\}$ with

$$\alpha^A = (p_i)_{i \in A} \cup \{Q_A\}, \qquad \beta^A = (\hat{p}_i)_{i \in A} \cup \{\hat{Q}_A\}.$$

The unique maximizer is therefore

$$g_i^* = \frac{p_i}{\hat{p}_i} \quad (i \in A), \qquad g_0^* = \frac{Q_A}{\hat{Q}_A}.$$

Since $f_i = \hat{p}_i(g_i - g_0)$ for $i \in A$, this gives (1). Equation (2) is just $r = g_0^*$. Substituting the optimizer into the objective yields (3). Finally,

$$f_i^A = \hat{p}_i \Big(\frac{p_i}{\hat{p}_i} - \frac{Q_A}{\hat{Q}_A} \Big),$$

which is equivalent to (4). □

Remark 3.3. Proposition 3.2 does more than solve the interior problem on a prescribed support. It reveals the natural candidate for the global optimum. On active states, the optimal wealth is the state-price likelihood ratio $p_i/\hat{p}_i = s_i$; on excluded states, the wealth is a common reserve level $Q_A/\hat{Q}_A$. The global proof below simply verifies that the corresponding wealth profile dominates every feasible competitor.

4 Global optimality by verification

The support can be identified without any iterative support construction. The right candidate is the maximal admissible prefix.

Definition 4.1. A prefix A_m is *admissible* if either $m = 0$, or $m \geq 1$ and

$$s_i > \rho(A_m) \qquad (1 \leq i \leq m).$$

Let m^* be the largest index for which A_{m^*} is admissible. Define

$$A^* := A_{m^*}, \qquad c := \rho(A^*) = \frac{Q_{A^*}}{\hat{Q}_{A^*}}.$$

For $m^* = 0$, interpret $A^* = \varnothing$ and $c := 1$.

Remark 4.2. This definition is natural for two reasons. First, admissibility is exactly the positivity condition from Proposition 3.2. Second, if A_m is admissible and $s_{m+1} > \rho(A_m)$, then A_{m+1} is admissible as well, because adjoining outcome $m + 1$ lowers the reserve ratio. Thus the largest admissible prefix is well defined.

Theorem 4.3 (Verification theorem for full Kelly)**.** *Let A^* be the largest admissible prefix and let $c = Q_{A^*}/\hat{Q}_{A^*}$. Define the candidate portfolio by*

$$z^* := c, \qquad f_i^* := p_i - c\hat{p}_i \quad (i \in A^*), \qquad f_j^* := 0 \quad (j \notin A^*),$$

where $z^* := 1 - \sum_i f_i^*$. Its terminal wealth profile is

$$W_i^* = \begin{cases} \dfrac{p_i}{\hat{p}_i} = s_i, & i \in A^*, \\ c, & i \notin A^*. \end{cases}$$

Then for every feasible portfolio f,

$$G(f) \le G(f^*).$$

Hence f^* is the unique Kelly-optimal portfolio. In particular, the active set is exactly A^*.

Proof. Let f be any feasible portfolio, and write

$$z := 1 - \sum_{i=1}^n f_i, \qquad W_i := z + \frac{f_i}{\hat{p}_i}.$$

By construction of A^*, the candidate portfolio is feasible: for $i \in A^*$ we have $s_i > c$, so $f_i^* = \hat{p}_i(s_i - c) > 0$, while $z^* = c \ge 0$.

We now compare $W = (W_i)$ to $W^* = (W_i^*)$. Since $\log x \le x - 1$ for $x > 0$,

$$G(f) - G(f^*) = \sum_{i=1}^n p_i \log \frac{W_i}{W_i^*} \le \sum_{i=1}^n p_i \Big(\frac{W_i}{W_i^*} - 1 \Big).$$

Thus it suffices to show that

$$\sum_{i=1}^n p_i \frac{W_i}{W_i^*} \le 1. \tag{5}$$

Expanding the left-hand side and using $W_i^* = s_i$ on A^* and $W_j^* = c$ on the complement gives

$$\sum_{i=1}^n p_i \frac{W_i}{W_i^*} = \sum_{i \in A^*} p_i \frac{W_i}{s_i} + \sum_{j \notin A^*} p_j \frac{W_j}{c} = \sum_{i \in A^*} \hat{p}_i W_i + \frac{1}{c} \sum_{j \notin A^*} p_j W_j.$$

Now insert $W_i = z + f_i/\hat{p}_i$:

$$\sum_{i \in A^*} \hat{p}_i W_i = z \hat{P}_{A^*} + \sum_{i \in A^*} f_i,$$

and

$$\frac{1}{c} \sum_{j \notin A^*} p_j W_j = \frac{z Q_{A^*}}{c} + \sum_{j \notin A^*} \frac{s_j}{c} f_j = z \hat{Q}_{A^*} + \sum_{j \notin A^*} \frac{s_j}{c} f_j,$$

because $Q_{A^*}/c = \hat{Q}_{A^*}$. Therefore

$$\sum_{i=1}^n p_i \frac{W_i}{W_i^*} = z(\hat{P}_{A^*} + \hat{Q}_{A^*}) + \sum_{i \in A^*} f_i + \sum_{j \notin A^*} \frac{s_j}{c} f_j = z + \sum_{i \in A^*} f_i + \sum_{j \notin A^*} \frac{s_j}{c} f_j.$$

Since $z + \sum_i f_i = 1$, this becomes

$$\sum_{i=1}^n p_i \frac{W_i}{W_i^*} = 1 + \sum_{j \notin A^*} \Big(\frac{s_j}{c} - 1 \Big) f_j. \tag{6}$$

By maximality of the admissible prefix, every excluded outcome satisfies $s_j \le c$. Since $f_j \ge 0$, each summand on the right-hand side of (6) is nonpositive. Hence (5) holds, and therefore $G(f) \le G(f^*)$.

Finally, strict concavity of G on the feasible simplex implies uniqueness. Since $f_j^* = 0$ for $j \notin A^*$ and $f_i^* > 0$ for $i \in A^*$, the active set is exactly A^*. □

Corollary 4.4 (Stopping rule and prefix scan)**.** *The optimal active set is the unique prefix A_m satisfying*

$$s_m > \rho(A_m) \ge s_{m+1}, \tag{7}$$

with the conventions $s_0 := +\infty$ and $s_{n+1} := -\infty$. Equivalently, after sorting by $s_i = p_i d_i$, one may scan forward and retain the largest admissible prefix.

Proof. By definition, $A^* = A_{m^*}$ is admissible, so $s_m > \rho(A_m)$ for all active $m \le m^*$. If $m^* < n$ and $s_{m^*+1} > \rho(A^*)$, then adjoining outcome $m^* + 1$ would produce another admissible prefix, contradicting maximality. Hence $\rho(A^*) \ge s_{m^*+1}$. Uniqueness follows from the ordering of the s_i. □

Remark 4.5 (Alternative Kullback–Leibler proof)*.* The verification theorem is the shortest global proof, but the support rule also admits a complementary one-step refinement interpretation. Suppose A is admissible and $k \notin A$ satisfies $s_k > \rho(A)$. Then $B := A \cup \{k\}$ is admissible as well, and Proposition 3.2 applies to both A and B. A direct subtraction yields

$$V(B) - V(A) = Q_A\, D_{\mathrm{KL}}\left(\left(\frac{p_k}{Q_A}, 1 - \frac{p_k}{Q_A}\right) \middle\| \left(\frac{\hat{p}_k}{\hat{Q}_A}, 1 - \frac{\hat{p}_k}{\hat{Q}_A}\right)\right).$$

Thus adjoining a conditionally favorable outcome increases value by a binary relative entropy. This KL identity is conceptually attractive, but the verification argument above is shorter and avoids any step-by-step support construction.

5 Interpretation

The proof above packages the multinomial Kelly rule into three closely related ideas.

First, on a fixed support, the problem is naturally a state-price problem for terminal wealth rather than a stake-allocation problem. Lemma 3.1 says that the expected-log-optimal Arrow–Debreu payoff is the likelihood ratio between the bettor's subjective law and the relevant state-price law. Proposition 3.2 then shows that, after coarse-graining all inactive outcomes into a reserve bucket, the optimized value is exactly the relative entropy between the corresponding coarse-grained laws.

Second, the global verification proof shows that the same likelihood-ratio wealth profile is already visible before the support is proved. The active states should have wealth $p_i/\hat{p}_i$, while the excluded states should share a common reserve level c. Once this candidate is postulated, the inequality $\log x \le x - 1$ reduces optimality to the single identity (6). The threshold condition $s_j \le c$ for excluded outcomes appears there as the exact penalty term for placing weight on an unfavorable state.

Third, the support rule itself has two equally natural readings. In the verification proof it appears as the maximal admissible prefix. In the alternative refinement argument of Remark 4.5, it appears as the condition for a positive KL gain when one outcome is split off from the reserve bucket. These are two views of the same information-theoretic structure.

In this way the classical multinomial Kelly rule becomes an exercise in relative entropy and state prices. The proof is elementary, but it clarifies why the explicit formula has the shape it does: the Kelly bettor is choosing the Arrow–Debreu payoff profile that maximizes expected log utility, and the reserve bucket enforces the correct threshold for support selection.

Acknowledgment of scope

The formula and support criterion proved here are classical. The point of the note is expository: it emphasizes that the single-event multinomial Kelly rule is naturally governed by relative entropy, state prices, and a simple verification principle.

References

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